

IIT JAM 2026 Official Paper

Category: CBT Era (2015–2026) • Official Mock Test Series

TOTAL QUESTIONS

60

TOTAL MARKS

100 M

TEST DURATION

180 Min

PAPER PATTERN

MCQ • MSQ • NAT
(60 Ques)

Section / Type	Questions	Marking Scheme	Negative Marking	Format / Details
Multiple Choice (MCQ)	30	+1 / +2 Marks	−1/3 / −2/3 Marks	4 Options (1 Correct)
Multiple Select (MSQ)	10	+2 Marks	Nil (0)	4 Options (1+ Correct)
Numerical Answer (NAT)	20	+1 / +2 Marks	Nil (0)	Real Decimal Value

IMPORTANT INSTRUCTIONS FOR CANDIDATES

- Authentic Examination PYQs:** This mock paper contains verified official IIT JAM Mathematics questions and high-resolution problem statements.
- Section A (MCQ):** Each question has four choices (A), (B), (C), (D), out of which **only one** is correct. Wrong answers carry negative penalty (1/3 of positive marks).
- Section B (MSQ):** Each question has four choices (A), (B), (C), (D), out of which **one or more than one** choice(s) may be correct. Full marks are awarded only if all correct choices and zero incorrect choices are chosen. No partial marks and no negative marks.
- Section C (NAT):** Numerical answers are real numbers entered via virtual keyboard. Full credit is awarded if the numerical answer falls within the official accepted interval. No negative marking.
- Master Answer Key:** Complete official answers and acceptance bounds are provided at the end of this document.
- Online Interactive Simulator:** Practice this test with virtual calculator, timers, and instant scoring analytics at vaibhavgeometer.github.io.

IIT JAM Mathematics Mock Test Series • Curated by Vaibhav Geometer

Question 1 (JAM 2026 • Q1 • ID: JAM_2026_Q1)

MCQ

+1 M

Neg: -0.33

Q.1	For each positive integer n , let $x_n = 1 - (-1)^n + \frac{1}{n}$ and $y_n = \left(1 + \frac{1}{2n}\right)^{3n}$. Which ONE of the following statements about the sequences (x_n) and (y_n) is TRUE?
(A)	(x_n) and (y_n) are convergent.
(B)	(x_n) is not convergent and (y_n) is not convergent.
(C)	(x_n) is convergent and (y_n) is not convergent.
(D)	(x_n) is not convergent and (y_n) is convergent.

Question 2 (JAM 2026 • Q2 • ID: JAM_2026_Q2)

MCQ

+1 M

Neg: -0.33

Q.2	For each positive integer n , define $x_n = (-1)^n$. Which ONE of the following statements about the sequence (x_n) is FALSE?
(A)	There exists $\epsilon > 0$ such that $ x_n - 1 < \epsilon$ for all positive integers n .
(B)	There exists $\epsilon > 0$ such that for all $M > 0$ there exists a positive integer $N > M$ for which $ x_N - 1 > \epsilon$.
(C)	For all $\epsilon > 0$ and $M > 0$ there exists a positive integer N such that $N > M$ and $ x_N - 1 < \epsilon$.
(D)	For all $\epsilon > 0$ and $M > 0$ there exists a positive integer N such that $N > M$ and $ x_N - 1 > \epsilon$.

Question 3 (JAM 2026 • Q3 • ID: JAM_2026_Q3)

MCQ

+1 M

Neg: -0.33

Q.3	Let $f: (1, 4) \rightarrow \mathbb{R}$ be a differentiable function such that $f'(x) = (f(x) - \pi x)^2 + \pi \quad \text{for all } x \in (1, 4).$ Which ONE of the following is a possible value of $f(3) - f(2)$?
(A)	$\pi + \frac{1}{6}$
(B)	$\pi - \frac{1}{6}$
(C)	$\frac{\pi}{2} + \frac{1}{3}$
(D)	$\frac{\pi}{2} + \frac{1}{2}$

Question 4 (JAM 2026 • Q4 • ID: JAM_2026_Q4)

MCQ

+1 M

Neg: -0.33

Q.4	The general solution of the differential equation $x \sin\left(\frac{y}{x}\right) \frac{dy}{dx} = y \sin\left(\frac{y}{x}\right) + \frac{x}{2}, \quad x > 0$ is given by _____.
(A)	$\cos\left(\frac{x}{y}\right) + \log_e(x^2) = K$, where K is an arbitrary constant
(B)	$\cos\left(\frac{y}{x}\right) + \log_e(x^2) = K$, where K is an arbitrary constant
(C)	$\cos\left(\frac{x}{y}\right) + \log_e(\sqrt{x}) = K$, where K is an arbitrary constant
(D)	$\cos\left(\frac{y}{x}\right) + \log_e(\sqrt{x}) = K$, where K is an arbitrary constant

Question 5 (JAM 2026 • Q5 • ID: JAM_2026_Q5)

MCQ

+1 M

Neg: -0.33

Q.5	Let $f: \mathbb{R}^3 \rightarrow \mathbb{R}$ be a continuous function. Then $\int_0^2 \int_0^x \int_0^y f(x, y, z) dz dy dx = \underline{\hspace{2cm}}.$
(A)	$\int_0^2 \int_0^y \int_0^2 f(x, y, z) dx dz dy$
(B)	$\int_0^2 \int_0^x \int_z^2 f(x, y, z) dy dz dx$
(C)	$\int_0^2 \int_y^2 \int_0^x f(x, y, z) dz dx dy$
(D)	$\int_0^2 \int_z^2 \int_0^x f(x, y, z) dy dx dz$

Question 6 (JAM 2026 • Q6 • ID: JAM_2026_Q6)

MCQ

+1 M

Neg: -0.33

Q.6	Let $f: (1, 2) \rightarrow \mathbb{R}$ be a continuous function satisfying $\lim_{x \rightarrow 1^+} f(x) = 3 \text{ and } \lim_{x \rightarrow 2^-} f(x) = 3.$ Then which ONE of the following is necessarily TRUE?
(A)	f is bounded and f has a maximum or a minimum but not both.
(B)	f is bounded and f has a maximum or a minimum or both.
(C)	f is bounded and f has neither a maximum nor a minimum.
(D)	f is unbounded and f has either no maximum or no minimum.

Question 7 (JAM 2026 • Q7 • ID: JAM_2026_Q7)

MCQ

+1 M

Neg: -0.33

Q.7	Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a twice differentiable function such that $f(n) = 1$ for all $n \in \mathbb{Z}$. Which ONE of the following statements is necessarily TRUE?
(A)	$f'(n) = 0$ for all $n \in \mathbb{Z}$.
(B)	$f'(x) = 0$ for infinitely many $x \in \mathbb{R} \setminus \mathbb{Z}$.
(C)	$f''(n) = 0$ for all $n \in \mathbb{Z}$.
(D)	$f''(x) = 0$ for infinitely many $x \in \mathbb{R} \setminus \mathbb{Z}$.

Question 8 (JAM 2026 • Q8 • ID: JAM_2026_Q8)

MCQ

+1 M

Neg: -0.33

Q.8	Let A be an invertible 5×5 real matrix. Let B be the matrix obtained by interchanging the second and third columns of A and then adding 3 times the third column to the fifth column. Which ONE of the following statements is TRUE?
(A)	B^{-1} is obtained by interchanging the second and third rows of A^{-1} , and then adding 3 times the third row to the fifth row.
(B)	B^{-1} is obtained by interchanging the second and third rows of A^{-1} , and then adding -3 times the fifth row to the third row.
(C)	B^{-1} is obtained by adding -3 times the third row to the fifth row, and then interchanging the second and third rows of A^{-1} .
(D)	B^{-1} is obtained by adding 3 times the fifth row to the third row, and then interchanging the second and third rows of A^{-1} .

Question 9 (JAM 2026 • Q9 • ID: JAM_2026_Q9)

MCQ

+1 M

Neg: -0.33

Q.9	Which ONE of the following statements is FALSE?
(A)	S_3 is isomorphic to a subgroup of S_4 .
(B)	$\mathbb{Z}_3$ is isomorphic to a subgroup of S_4 .
(C)	S_3 is isomorphic to a quotient group of S_4 .
(D)	$\mathbb{Z}_6$ is isomorphic to a quotient group of S_4 .

Question 10 (JAM 2026 • Q10 • ID: JAM_2026_Q10)

MCQ

+1 M

Neg: -0.33

Q.10	Let T be the triangular region in the plane with vertices at $(0,0)$, $(1,0)$ and $(1,1)$. Let f be a function defined from T to $\mathbb{R}$ given in polar coordinates. Then the double integral of f over T is equal to _____.
(A)	$\int_0^{\sqrt{2}} \left(\int_0^{\pi/4} f \, d\theta \right) r \, dr + \int_1^{\sqrt{2}} \left(\int_0^{\cos^{-1}(1/r)} f \, d\theta \right) r \, dr$
(B)	$\int_0^{\sqrt{2}} \left(\int_0^{\pi/4} f \, d\theta \right) r \, dr - \int_1^{\sqrt{2}} \left(\int_0^{\cos^{-1}(1/r)} f \, d\theta \right) r \, dr$
(C)	$\int_0^{\sqrt{2}} \left(\int_0^{\pi/4} f \, d\theta \right) r \, dr + \int_1^{\sqrt{2}} \left(\int_0^{\sin^{-1}(1/r)} f \, d\theta \right) r \, dr$
(D)	$\int_0^{\sqrt{2}} \left(\int_0^{\pi/4} f \, d\theta \right) r \, dr - \int_1^{\sqrt{2}} \left(\int_0^{\sin^{-1}(1/r)} f \, d\theta \right) r \, dr$

Question 11 (JAM 2026 • Q11 • ID: JAM_2026_Q11)

MCQ

+2 M

Neg: -0.67

Q.11	Given a sequence of real numbers (a_n), define $b_n = \begin{cases} a_n, & \text{if } a_n \geq 0 \\ 0, & \text{if } a_n < 0 \end{cases} \text{ and } c_n = \begin{cases} a_n, & \text{if } a_n < 0 \\ 0, & \text{if } a_n \geq 0 \end{cases}$ for each positive integer n. Which ONE of the following statements is TRUE?
(A)	If $\sum_{n=1}^{\infty} a_n$ does not converge, then $\sum_{n=1}^{\infty} b_n$ does not converge.
(B)	If $\sum_{n=1}^{\infty} a_n$ converges but $\sum_{n=1}^{\infty} a_n $ does not converge, then $\sum_{n=1}^{\infty} b_n$ converges.
(C)	If $\sum_{n=1}^{\infty} a_n$ converges but $\sum_{n=1}^{\infty} a_n $ does not converge, then $\sum_{n=1}^{\infty} b_n$ does not converge.
(D)	If $\sum_{n=1}^{\infty} a_n$ converges, then $\sum_{n=1}^{\infty} b_n$ converges or $\sum_{n=1}^{\infty} c_n$ converges.

Question 12 (JAM 2026 • Q12 • ID: JAM_2026_Q12)

MCQ

+2 M

Neg: -0.67

Q.12	For which one of the following pairs of values of a and b , does the series $\sum_{k=1}^{\infty} \frac{(3x)^k}{2^{\sqrt{k}}(1-x)^k}$ converge for all $x \in (a, b)$?
(A)	$a = \frac{-1}{4}$ and $b = \frac{1}{2}$
(B)	$a = -1$ and $b = \frac{1}{5}$
(C)	$a = \frac{-1}{2}$ and $b = \frac{1}{4}$
(D)	$a = \frac{-1}{4}$ and $b = \frac{1}{3}$

Question 13 (JAM 2026 • Q13 • ID: JAM_2026_Q13)

MCQ

+2 M

Neg: -0.67

Q.13	Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be a function that has derivatives of all orders. Which ONE of the following statements is FALSE?
(A)	If f is a polynomial in x of degree at most 5, then $f^{(6)}(x) = 0$ for all $x \in \mathbb{R}$.
(B)	If $f^{(6)}(x) = 0$ for all $x \in \mathbb{R}$, then f is a polynomial in x of degree at most 5.
(C)	If $f^{(k)}(a) = 0$ for all $1 \leq k \leq 5$ and $f^{(6)}(a) > 0$ for some $a \in \mathbb{R}$, then $f(x)$ has a local minimum at a .
(D)	If $f^{(k)}(a) = 0$ for all $1 \leq k \leq 6$ and $f^{(7)}(a) < 0$ for some $a \in \mathbb{R}$, then $f(x)$ has a local maximum at a .

Question 14 (JAM 2026 • Q14 • ID: JAM_2026_Q14)

MCQ

+2 M

Neg: -0.67

Q.14	Let A be a 5×3 real matrix of rank 2 and b be a non-zero 5×1 real column vector. Suppose $\begin{pmatrix} 1 \\ 2 \\ 3 \end{pmatrix}$ and $\begin{pmatrix} 4 \\ 5 \\ 6 \end{pmatrix}$ are two solutions of the system of linear equations $Ax = b$. Which ONE of the following is also a solution of $Ax = b$?
(A)	$\begin{pmatrix} 3 \\ 3 \\ 3 \end{pmatrix}$
(B)	$\begin{pmatrix} 5 \\ 7 \\ 9 \end{pmatrix}$
(C)	$\begin{pmatrix} 5 \\ 6 \\ 9 \end{pmatrix}$
(D)	$\begin{pmatrix} 7 \\ 8 \\ 9 \end{pmatrix}$

Question 15 (JAM 2026 • Q15 • ID: JAM_2026_Q15)

MCQ

+2 M

Neg: -0.67

Q.15	Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function given by $f(x, y) = (x^2 - 1)^2 + (y^2 - 1)^2 \text{ for all } (x, y) \in \mathbb{R}^2.$ Which ONE of the following statements is TRUE?
(A)	f has local maxima at exactly two points.
(B)	f has local minima at exactly two points.
(C)	f has local minima at exactly three points.
(D)	f has exactly four saddle points.

Question 16 (JAM 2026 • Q16 • ID: JAM_2026_Q16)

MCQ

+2 M

Neg: -0.67

Q.16	The orthogonal trajectories of the family of curves $y = -3x - 3 + me^x \quad \text{for a real parameter } m$ are given by _____.
(A)	$x = \frac{y}{3} - \frac{1}{9} + k e^{-3y}$, k is a real parameter
(B)	$x = \frac{y}{3} - \frac{1}{9} + k e^{3y}$, k is a real parameter
(C)	$x = \frac{-y}{3} + \frac{1}{9} + k e^{-3y}$, k is a real parameter
(D)	$x = \frac{-y}{3} + \frac{1}{9} + k e^{3y}$, k is a real parameter

Question 17 (JAM 2026 • Q17 • ID: JAM_2026_Q17)

MCQ

+2 M

Neg: -0.67

Q.17

Let $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be the function defined by

$$f(x, y) = \begin{cases} \frac{5x^2y - 4y^3}{x^2 + y^2}, & (x, y) \neq (0, 0) \\ 0, & (x, y) = (0, 0) \end{cases}$$

Which ONE of the following statements is FALSE?

(A) f is continuous at $(0, 0)$ (B) $\frac{\partial f}{\partial x}$ at $(0, 1) = 0$ (C) $\frac{\partial f}{\partial y}$ at $(1, 0) = 5$ (D) $\frac{\partial f}{\partial y}$ is continuous at $(0, 0)$

Question 18 (JAM 2026 • Q18 • ID: JAM_2026_Q18)

MCQ

+2 M

Neg: -0.67

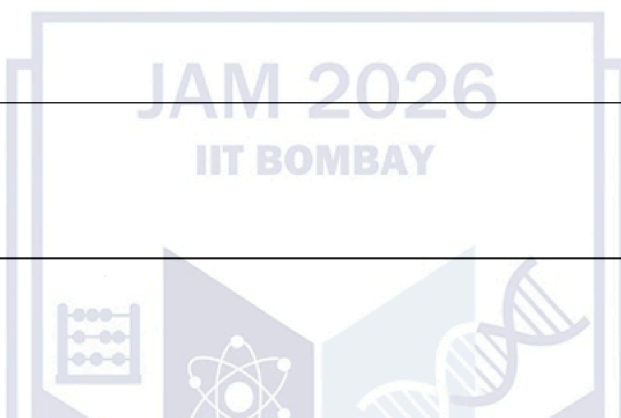
Q.18

Let $\phi: \left(-\frac{\pi}{2}, \frac{\pi}{2}\right) \rightarrow \mathbb{R}$ be the solution of the differential equation

$$(\cos x) \frac{dy}{dx} - y = 2y^2(\sin x - 1) \cos x$$

satisfying $\phi(0) = \frac{1}{2}$. Then $\phi\left(\frac{\pi}{4}\right) = \underline{\hspace{2cm}}$.

(A) 2

(B) $\frac{3 + 2\sqrt{2}}{2}$ (C) $\frac{1}{2}$ (D) $\frac{1}{\sqrt{2}}$ 

Question 19 (JAM 2026 • Q19 • ID: JAM_2026_Q19)

MCQ

+2 M

Neg: -0.67

Q.19	The value of the following expression $\lim_{x \rightarrow \infty} \int_0^x e^{-t} \sin t dt$ is ____.
(A)	$\frac{e^\pi + 1}{2(e^\pi - 1)}$
(B)	$\frac{e^\pi - 1}{2(e^\pi + 1)}$
(C)	$\frac{2(e^\pi + 1)}{e^\pi - 1}$
(D)	$\frac{2(e^\pi - 1)}{e^\pi + 1}$

Question 20 (JAM 2026 • Q20 • ID: JAM_2026_Q20)

MCQ

+2 M

Neg: -0.67

Q.20	The semi-circle $y = \sqrt{6x - 5 - x^2}$ is rotated clockwise by the angle $\frac{\pi}{2}$ in the plane about the origin. What is the area of the planar region traced out by the semi-circle?
(A)	6π
(B)	4π
(C)	2π
(D)	π

Question 21 (JAM 2026 • Q21 • ID: JAM_2026_Q21)

MCQ

+2 M

Neg: -0.67

Q.21	Let $f: [1, 9] \rightarrow \mathbb{R}$ be a non-constant continuous function such that $f(1) = f(9)$. Which ONE of the following statements is necessarily TRUE?
(A)	There exists $c \in [4, 8]$ such that $f(c) = f(c + 1)$.
(B)	There exists $c \in [3, 7]$ such that $f(c) = f(c + 2)$.
(C)	There exists $c \in [2, 6]$ such that $f(c) = f(c + 3)$.
(D)	There exists $c \in [1, 5]$ such that $f(c) = f(c + 4)$.

Question 22 (JAM 2026 • Q22 • ID: JAM_2026_Q22)

MCQ

+2 M

Neg: -0.67

Q.22	Let S be the solid of intersection of two solid spheres S_1, S_2 given by $S_1: x^2 + y^2 + (z - 1)^2 \leq 4 \quad \text{and} \quad S_2: x^2 + y^2 + (z + 1)^2 \leq 4 .$ Which ONE of the following iterated integrals expresses the volume of S?
(A)	$8 \int_0^{\sqrt{3}} \int_0^{\sqrt{3-x^2}} (\sqrt{4-x^2-y^2} - 1) dy dx$
(B)	$8 \int_0^2 \int_0^{\sqrt{4-x^2}} (\sqrt{4-x^2-y^2} - 1) dy dx$
(C)	$2 \int_0^2 \int_0^{\sqrt{4-x^2}} (\sqrt{4-x^2-y^2} - 1) dy dx$
(D)	$2 \int_0^{\sqrt{3}} \int_0^{\sqrt{3-x^2}} (\sqrt{4-x^2-y^2} - 1) dy dx$

Question 23 (JAM 2026 • Q23 • ID: JAM_2026_Q23)

MCQ

+2 M

Neg: -0.67

Q.23	Consider the differential equation $\frac{d^2y}{dx^2} + 3 \frac{dy}{dx} + 2y = \sin(e^x).$ Which ONE of the following is a particular solution of this differential equation?
(A)	$-e^{-x} \sin(e^x)$
(B)	$-e^{-2x} \sin(e^x)$
(C)	$-e^{-x} (\sin(e^x) + \cos(e^x))$
(D)	$-e^{-2x} (\sin(e^x) + \cos(e^x))$

Question 24 (JAM 2026 • Q24 • ID: JAM_2026_Q24)

MCQ

+2 M

Neg: -0.67

Q.24

Let e_1, e_2, e_3, e_4, e_5 denote the standard basis vectors of the real vector space $\mathbb{R}^5$. Let V be the subspace of $\mathbb{R}^5$ spanned by the vectors $e_1 + e_2, e_2 + e_3, e_3 + e_4 + e_5$. Consider the real vector space $M_{2,5}(\mathbb{R})$. Let

$$S = \{P \in M_{2,5}(\mathbb{R}) \mid \text{nullspace}(P) = V\}.$$

Which ONE of the following statements is TRUE?

(A) S is the empty set.

(B) $S = \{0\}$.

(C) S is non-empty and S is not a subspace of $M_{2,5}(\mathbb{R})$.

(D) S is a nonzero subspace of $M_{2,5}(\mathbb{R})$.

Question 25 (JAM 2026 • Q25 • ID: JAM_2026_Q25)

MCQ

+2 M

Neg: -0.67

Q.25	Let P be a nonzero complex $n \times n$ matrix such that $v^* P v \geq 0$ for all column vectors v in $\mathbb{C}^n$. Which ONE of the following statements is necessarily TRUE?
(A)	All eigenvalues of P are negative real numbers.
(B)	If v_1, v_2 are column vectors in $\mathbb{C}^n$ such that $P v_1 = v_2$ and $P v_2 = v_1$, then $v_1 = v_2$.
(C)	$v^* P v \neq 0$ for all nonzero column vectors v in $\mathbb{C}^n$.
(D)	All eigenvalues of P^* are negative real numbers.

Question 26 (JAM 2026 • Q26 • ID: JAM_2026_Q26)

MCQ

+2 M

Neg: -0.67

Q.26	Let $f: G \rightarrow \mathbb{Z}$ be a surjective group homomorphism from an abelian group G to the additive group of integers $\mathbb{Z}$. Let K denote the kernel of f. Which ONE of the following statements is FALSE?
(A)	There exists a group homomorphism $\phi: \mathbb{Z} \rightarrow G$ such that $f \circ \phi$ is the identity homomorphism.
(B)	K is isomorphic to a quotient group of G .
(C)	For all non-zero homomorphisms $\alpha: \mathbb{Z} \rightarrow G$ and $\beta: G \rightarrow K$, the composition $\beta \circ \alpha$ is non-zero.
(D)	G has a subgroup isomorphic to $\mathbb{Z}$.

Question 27 (JAM 2026 • Q27 • ID: JAM_2026_Q27)

MCQ

+2 M

Neg: -0.67

Q.27	Which ONE of the following statements is FALSE?
(A)	There is a surjective group homomorphism from the additive group of rational numbers to the multiplicative group of all complex roots of unity.
(B)	The multiplicative group of complex numbers of modulus one is isomorphic to a quotient group of the additive group of real numbers.
(C)	Any group homomorphism from the multiplicative group of nonzero complex numbers into the group of all invertible 2×2 matrices with real entries has nontrivial kernel.
(D)	There exists a group homomorphism from the symmetric group on n symbols into the multiplicative group of all invertible $n \times n$ matrices with real entries, which has a trivial kernel.

Question 28 (JAM 2026 • Q28 • ID: JAM_2026_Q28)

MCQ

+2 M

Neg: -0.67

Q.28	For any two points $P, Q \in \mathbb{R}^3$, let $\text{vect}(P, Q)$ denote the vector from the point P to the point Q and let $d(P, Q)$ denote the length of $\text{vect}(P, Q)$. Let $F: \mathbb{R}^3 \rightarrow \mathbb{R}^3$ be a linear transformation such that $d(F(P), F(Q)) = d(P, Q) \quad \text{for all } P, Q \in \mathbb{R}^3.$ Which ONE of the following statements is FALSE ?
(A)	F is an injective map.
(B)	F is a surjective map.
(C)	For any four points P, Q, R, S in $\mathbb{R}^3$, we have $\text{vect}(F(P), F(Q)) \cdot \text{vect}(F(R), F(S)) = \text{vect}(P, Q) \cdot \text{vect}(R, S),$ where $\cdot$ denotes the usual dot product in $\mathbb{R}^3$.
(D)	For any four points P, Q, R, S in $\mathbb{R}^3$, we have $\text{vect}(F(P), F(Q)) \times \text{vect}(F(R), F(S)) = \text{vect}(P, Q) \times \text{vect}(R, S),$ where $\times$ denotes the usual cross product in $\mathbb{R}^3$.

Question 29 (JAM 2026 • Q29 • ID: JAM_2026_Q29)

MCQ

+2 M

Neg: -0.67

Q.29

For each positive integer n , let

$$x_n = 1 + \frac{1}{2} + \cdots + \frac{1}{n} - \log_e n$$

and

$$y_n = \int_1^n \frac{\cos t}{t^2} dt .$$

Which ONE of the following statements about the sequences (x_n) and (y_n) is TRUE?

- (A) (x_n) and (y_n) are convergent.
- (B) (x_n) is convergent and (y_n) is not convergent.
- (C) (x_n) is not convergent and (y_n) is convergent.
- (D) (x_n) is not convergent and (y_n) is not convergent.

Question 30 (JAM 2026 • Q30 • ID: JAM_2026_Q30)

MCQ

+2 M

Neg: -0.67

Q.30	Let G be the power set of $\{1,2,3,4\}$. Let the binary operation Δ on G be $A \Delta B = (A \setminus B) \cup (B \setminus A).$ Which ONE of the following statements is TRUE?
(A)	$\{1\}$ is an identity element of (G, Δ) .
(B)	(G, Δ) is an abelian group but not cyclic.
(C)	(G, Δ) is a group and has an element of order 4.
(D)	(G, Δ) is a group and has an element of order 8.

Question 31 (JAM 2026 • Q31 • ID: JAM_2026_Q31)

MSQ

+2 M

Neg: Nil (0)

Q.31	Which of the following functions $f: \mathbb{R} \rightarrow \mathbb{R}$ has/have a local minimum at $x = 0$?
(A)	$f(x) = \sin x $
(B)	$f(x) = \sin x + \frac{x^3}{6}$
(C)	$f(x) = x^4 + x^2 + 3$
(D)	$f(x) = \min \{x - [x], 1 - x + [x]\}$

Question 32 (JAM 2026 • Q32 • ID: JAM_2026_Q32)

MSQ

+2 M

Neg: Nil (0)

Q.32	Which of the following statements is/are TRUE?
(A)	There exists a monotone sequence that does not converge but has a convergent subsequence.
(B)	There exists a sequence that has a bounded subsequence but does not have any convergent subsequence.
(C)	There exists a sequence (x_n) such that given any positive integer m , (x_n) has a subsequence converging to m .
(D)	There exists a sequence (x_n) such that $(x_{n+1} - x_n)$ converges to 0 but (x_n) does not converge.

Question 33 (JAM 2026 • Q33 • ID: JAM_2026_Q33)

MSQ

+2 M

Neg: Nil (0)

Q.33	Which of the following functions is/are differentiable at $x = 1$?
(A)	$f(x) = x - 1 ^3$
(B)	$f(x) = x^2 - 1 $
(C)	$f(x) = \begin{cases} x^2 e^{-x^2}, & x \leq 1 \\ e^{-1}, & x > 1 \end{cases}$
(D)	$f(x) = [x]$

Question 34 (JAM 2026 • Q34 • ID: JAM_2026_Q34)

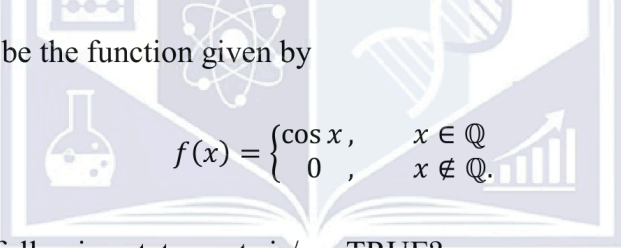
MSQ

+2 M

Neg: Nil (0)

Q.34

Let $f: \mathbb{R} \rightarrow \mathbb{R}$ be the function given by


$$f(x) = \begin{cases} \cos x, & x \in \mathbb{Q} \\ 0, & x \notin \mathbb{Q}. \end{cases}$$

Which of the following statements is/are TRUE?

(A) $f(x)$ is continuous at 0.(B) $f(x)$ is continuous at $\frac{\pi}{2}$.(C) $f(x)$ is Riemann integrable on $[0, 1]$ and $\int_0^1 f(x) dx = \sin 1$.(D) $f(x)$ is Riemann integrable on $[0, 1]$ and $\int_0^1 f(x) dx = 0$.

Question 35 (JAM 2026 • Q35 • ID: JAM_2026_Q35)

MSQ

+2 M

Neg: Nil (0)

Q.35	Consider the differential equation $(2y \cos x - xy \sin x) dx + 2x \cos x dy = 0 \quad \text{for } x \in \left(0, \frac{\pi}{4}\right).$ Which of the following is/are integrating factor(s) of the differential equation?
(A)	$\frac{1}{xy}$
(B)	xy
(C)	$\sec x$
(D)	$\sqrt{\sec x}$

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Question 36 (JAM 2026 • Q36 • ID: JAM_2026_Q36)

MSQ

+2 M

Neg: Nil (0)

Q.36

$$\text{Let } f(x, y) = \begin{cases} \frac{x^2 y}{1+x^2} \sin(1/x), & x \neq 0 \\ 0, & x = 0 \end{cases}$$

Which of the following statements is/are TRUE?

- (A) $\lim_{y \rightarrow 0} \lim_{x \rightarrow 0} f(x, y)$ exists.
- (B) $\frac{\partial f}{\partial x}$ is continuous at the point $(0, 1)$.
- (C) $\frac{\partial f}{\partial y}$ is continuous at the point $(0, 1)$.
- (D) $\lim_{(x, y) \rightarrow (0, 0)} \frac{f(x, y)}{\sqrt{x^2 + y^2}}$ does not exist.

Question 37 (JAM 2026 • Q37 • ID: JAM_2026_Q37)

MSQ

+2 M

Neg: Nil (0)

Q.37

$$\text{Let } P = \begin{bmatrix} 0 & 1 & 0 & 0 & 0 \\ 1 & 0 & 0 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 0 & 1 \\ 0 & 0 & 1 & 0 & 0 \end{bmatrix} \in M_5(\mathbb{C}).$$

Which of the following statements is/are TRUE?

- (A) $\text{nullity}(P - I) \geq 2$, where I is the 5×5 identity matrix.
- (B) P has 4 distinct eigenvalues in $\mathbb{C}$.
- (C) P has 3 distinct eigenvalues in $\mathbb{R}$.
- (D) If λ is an eigenvalue of P , then there exists a positive integer n such that $\lambda^n = 1$.

Question 38 (JAM 2026 • Q38 • ID: JAM_2026_Q38)

MSQ

+2 M

Neg: Nil (0)

Q.38	Let $\binom{n}{r}$ denote the number of ways of choosing r distinct objects out of n distinct objects. Which of the following statements is/are TRUE?
(A)	$\sum_{k=0}^6 (-1)^k \binom{13}{2k} = -64.$
(B)	$\sum_{k=0}^6 (-1)^k \binom{13}{2k+1} = 128.$
(C)	$\sum_{k=0}^6 (-1)^k \binom{13}{2k} = 64.$
(D)	$\sum_{k=0}^6 (-1)^k \binom{13}{2k+1} = -128.$

Question 39 (JAM 2026 • Q39 • ID: JAM_2026_Q39)

MSQ

+2 M

Neg: Nil (0)

Q.39	Consider matrices P of order 4×6 and Q of order 6×4 with real entries such that $PQ = 0$ and $QP = 0$. Which of the following statements is/are TRUE?
(A)	$\text{rangespace}(P) \subseteq \text{nullspace}(Q)$ and $\text{rangespace}(Q) \subseteq \text{nullspace}(P)$.
(B)	$\text{rank}(P) + \text{rank}(Q) \leq 4$.
(C)	If $\text{rangespace}(P) = \text{nullspace}(Q)$, then $\text{rank}(P) + \text{rank}(Q) = 4$.
(D)	$\text{rangespace}(Q) = \text{nullspace}(P)$.

Question 40 (JAM 2026 • Q40 • ID: JAM_2026_Q40)

MSQ

+2 M

Neg: Nil (0)

Q.40	Let V be the subset of $\mathbb{R}$ defined by $V = \left\{ \frac{a + b\sqrt{2}}{c + d\sqrt{2}} : a, b, c, d \in \mathbb{Q}, c^2 + d^2 \neq 0 \right\}.$ Which of the following statements is/are FALSE?
(A)	V is closed under the usual addition in $\mathbb{R}$.
(B)	V is a subspace of the real vector space $\mathbb{R}$.
(C)	V is a two dimensional subspace of the vector space $\mathbb{R}$ over $\mathbb{Q}$.
(D)	V is a four dimensional subspace of the vector space $\mathbb{R}$ over $\mathbb{Q}$.
Section C: Q.41 – Q.50 Carry ONE mark each.	

Question 41 (JAM 2026 • Q41 • ID: JAM_2026_Q41)

NAT

+1 M

Neg: Nil (0)

Q.41

The radius of convergence of the series $\sum_{n=1}^{\infty} \frac{(n!)^2}{(2n)!} (\log_e n)^{-1} x^n$ is _____
(rounded off to one decimal place).

Question 42 (JAM 2026 • Q42 • ID: JAM_2026_Q42)

NAT

+1 M

Neg: Nil (0)

Q.42

$\int_0^1 \left(\sum_{k=1}^{\infty} \frac{(\log_e 2)^k x^{k-1}}{(k-1)!} \right) dx = \text{_____}$ (rounded off to one decimal place).

Question 43 (JAM 2026 • Q43 • ID: JAM_2026_Q43)

NAT

+1 M

Neg: Nil (0)

Q.43

$\lim_{n \rightarrow \infty} \left(\frac{n^2+1}{\sqrt{n^6+1}} + \cdots + \frac{n^2+n}{\sqrt{n^6+n}} \right) = \text{_____}$ (rounded off to one decimal place).

Question 44 (JAM 2026 • Q44 • ID: JAM_2026_Q44)

NAT

+1 M

Neg: Nil (0)

Q.44

$\lim_{x \rightarrow 0} \left(\frac{1}{x} - \frac{1}{\sin x} + e^{\frac{1-\cos x}{x}} \right) = \text{_____}$ (rounded off to one decimal place).

Question 45 (JAM 2026 • Q45 • ID: JAM_2026_Q45)

NAT

+1 M

Neg: Nil (0)

Q.45

Let α and β be real numbers such that the differential equation

$$(y^3 + \alpha xy^4 - 5x + \cos 2y)dx + (3xy^2 + 20x^2y^3 + \beta x \sin 2y)dy = 0$$

is exact.

Then $\alpha + \beta = \text{_____}$ (rounded off to one decimal place).

Question 46 (JAM 2026 • Q46 • ID: JAM_2026_Q46)

NAT

+1 M

Neg: Nil (0)

Q.46

Let $z = \cos(4x + 5y)$, where $x = \frac{\pi}{2} + 2\theta$, $y = -\left(\frac{\pi}{4} + \theta\right)$.

Then $\frac{dz}{d\theta}\bigg|_{\theta=\frac{\pi}{4}} = \underline{\hspace{2cm}}$ (rounded off to one decimal place).

Q.47

Let C_n denote a cyclic group having n elements. If there is a surjective group homomorphism from C_n to C_{30} , then the total number of such distinct surjective homomorphisms is $\underline{\hspace{2cm}}$.

Question 48 (JAM 2026 • Q48 • ID: JAM_2026_Q48)

NAT

+1 M

Neg: Nil (0)

Q.48

$\int_0^3 (|x-1| - x[x]) dx = \underline{\hspace{2cm}}$ (rounded off to one decimal place).

Question 49 (JAM 2026 • Q49 • ID: JAM_2026_Q49)

NAT

+1 M

Neg: Nil (0)

Q.49

Let $A \in M_3(\mathbb{C})$. Suppose the column vector $v = \begin{pmatrix} \sqrt{5}i \\ 2i \\ x \end{pmatrix}$ in $\mathbb{C}^3$ belongs to the intersection of $\text{nullspace}(A)$ and $\text{rangespace}(A^T)$.
Then $|x| = \underline{\hspace{2cm}}$ (rounded off to one decimal place).

Question 50 (JAM 2026 • Q50 • ID: JAM_2026_Q50)

NAT

+1 M

Neg: Nil (0)

Q.50

Let P be a 5×5 real matrix with $\det(P) = 2$. Let Q be the matrix of cofactors of P . Then $\det(Q) = \underline{\hspace{2cm}}$ (rounded off to one decimal place).

Question 51 (JAM 2026 • Q51 • ID: JAM_2026_Q51)

NAT

+2 M

Neg: Nil (0)

Q.51

Let α, β and γ be fixed real numbers such that

$y(x) = C_1 e^{-x} + C_2 e^{2x} + \alpha x e^{-x}$, where C_1 and C_2 are arbitrary real constants,

is the general solution of the differential equation

$$\frac{d^2 y}{dx^2} + \beta \frac{dy}{dx} + \gamma y = -e^{-x}.$$

Then $\alpha(\beta + \gamma) = \underline{\hspace{2cm}}$ (rounded off to one decimal place).

Question 52 (JAM 2026 • Q52 • ID: JAM_2026_Q52)

NAT

+2 M

Neg: Nil (0)

Q.52

Let the function $f : \mathbb{R}^2 \rightarrow \mathbb{R}$ be defined by

$$f(x, y) = \begin{cases} 4x^2 \tan^{-1}\left(\frac{y}{2x}\right) + y^3 \tan^{-1}\left(\frac{x}{4y^2}\right), & xy \neq 0 \\ 0, & \text{otherwise} \end{cases}$$

Then $\left(\frac{\partial}{\partial y} \left(\frac{\partial f}{\partial x}\right)\right)$ at $(0,0)$ is _____ (rounded off to two decimal places).

Question 53 (JAM 2026 • Q53 • ID: JAM_2026_Q53)

NAT

+2 M

Neg: Nil (0)

Q.53

The double integral of $f(x, y) = x$ over the triangular region with vertices at $\left(-\frac{1}{2}, \frac{1}{2}\right)$, $(1, 2)$ and $(1, -1)$ is _____ (rounded off to one decimal place).

Question 54 (JAM 2026 • Q54 • ID: JAM_2026_Q54)

NAT

+2 M

Neg: Nil (0)

Q.54

Let

$$f(x) = \int_0^{e^{\sin x}} \int_0^{\log_e y} e^{-\left(\frac{1}{\sqrt{2}}\right)} (1 - t^2) dt dy.$$

Then $f'(\pi/4) =$ _____ (rounded off to one decimal place).JAM 2026
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Question 55 (JAM 2026 • Q55 • ID: JAM_2026_Q55)

NAT

+2 M

Neg: Nil (0)

Q.55

The volume of the tetrahedron bounded by the planes $x = 1, y = 2, z = 3$ and $12x + 8y + 6z = 70$ is _____ (rounded off to one decimal place).

Q.56

Let $M \in M_3(\mathbb{R})$. If

$$M \begin{pmatrix} \sqrt{15} \\ \sqrt{15} \\ 0 \end{pmatrix} = \begin{pmatrix} \sqrt{3} \\ \sqrt{2} \\ -5 \end{pmatrix}, \quad M \begin{pmatrix} 0 \\ 0 \\ \sqrt{30} \end{pmatrix} = \begin{pmatrix} \sqrt{15} \\ \sqrt{10} \\ \sqrt{5} \end{pmatrix}, \quad M \begin{pmatrix} -\sqrt{15} \\ \sqrt{15} \\ 0 \end{pmatrix} = \begin{pmatrix} \sqrt{12} \\ -\sqrt{18} \\ 0 \end{pmatrix}$$

then $\det(M) =$ _____ (rounded off to one decimal place).

Question 57 (JAM 2026 • Q57 • ID: JAM_2026_Q57)

NAT

+2 M

Neg: Nil (0)

Q.57

Let $a, b \in \mathbb{R}$. If the system of linear equations

$$x + 2y + 2z = 1$$

$$2x + 3y + z = 2$$

$$ax + 5y + bz = b$$

has infinitely many solutions, then $a + b =$ _____ (rounded off to one decimal place).

Question 58 (JAM 2026 • Q58 • ID: JAM_2026_Q58)

NAT

+2 M

Neg: Nil (0)

Q.58

A fruit shop has 4 different types of bananas. The number of ways in which 12 bananas can be bought with at least one banana from each type, is _____.

Question 59 (JAM 2026 • Q59 • ID: JAM_2026_Q59)

NAT

+2 M

Neg: Nil (0)

Q.59

Let A be a 3×3 real matrix such that given any column vector $x \in \mathbb{R}^3$, the column vector Ax is the reflection of x about the plane

$$\{(a, b, -a - b) : a, b \in \mathbb{R}\}.$$

Then the sum of the diagonal elements of A is _____ (rounded off to one decimal place).

Question 60 (JAM 2026 • Q60 • ID: JAM_2026_Q60)

NAT

+2 M

Neg: Nil (0)

Q.60

Let $\binom{n}{r}$ denote the number of ways of choosing r distinct objects out of n distinct objects. Then,

$$\binom{5}{0} + \binom{6}{1} + \binom{7}{2} + \binom{8}{3} + \binom{9}{4} + \binom{10}{5} + \binom{11}{6} = \text{_____}.$$

OFFICIAL MASTER ANSWER KEY & EVALUATION RUBRIC

IIT JAM 2026 Official Paper • All Official Answers & Acceptance Ranges

Q. No.	Type	Marks	Official Key
1	MCQ	+1	D
2	MCQ	+1	D
3	MCQ	+1	A
4	MCQ	+1	D
5	MCQ	+1	A
6	MCQ	+1	B
7	MCQ	+1	B
8	MCQ	+1	B
9	MCQ	+1	D
10	MCQ	+1	MTA
11	MCQ	+2	C
12	MCQ	+2	C
13	MCQ	+2	D
14	MCQ	+2	D
15	MCQ	+2	D
16	MCQ	+2	C
17	MCQ	+2	D
18	MCQ	+2	D
19	MCQ	+2	A
20	MCQ	+2	A

Q. No.	Type	Marks	Official Key
21	MCQ	+2	D
22	MCQ	+2	A
23	MCQ	+2	B
24	MCQ	+2	C
25	MCQ	+2	B
26	MCQ	+2	C
27	MCQ	+2	C
28	MCQ	+2	D
29	MCQ	+2	A
30	MCQ	+2	B
31	MSQ	+2	A;C;D
32	MSQ	+2	C;D
33	MSQ	+2	A;C
34	MSQ	+2	B
35	MSQ	+2	B;D
36	MSQ	+2	A;C
37	MSQ	+2	A;B;D
38	MSQ	+2	A
39	MSQ	+2	A;B;C
40	MSQ	+2	B;D

Q. No.	Type	Marks	Official Key
41	NAT	+1	4.0 to 4.0
42	NAT	+1	1.0 to 1.0
43	NAT	+1	1.0 to 1.0
44	NAT	+1	1.0 to 1.0
45	NAT	+1	8.0 to 8.0
46	NAT	+1	3.0 to 3.0
47	NAT	+1	8 to 8
48	NAT	+1	-4.0 to -4.0
49	NAT	+1	3.0 to 3.0
50	NAT	+1	16.0 to 16.0
51	NAT	+2	-1.0 to -1.0
52	NAT	+2	0.24 to 0.26
53	NAT	+2	1.1 to 1.3
54	NAT	+2	0.4 to 0.5
55	NAT	+2	4.0 to 4.0
56	NAT	+2	-1.0 to -1.0
57	NAT	+2	6.0 to 6.0
58	NAT	+2	165 to 165
59	NAT	+2	1.0 to 1.0
60	NAT	+2	924 to 924

SCORING METHODOLOGY & EVALUATION GUIDELINES

- **Multiple Choice Questions (MCQ):** Full marks for correct single choice. Negative marks deducted for each incorrect attempt (-1/3 for 1-mark questions, -2/3 for 2-mark questions). Unattempted questions award zero.
- **Multiple Select Questions (MSQ):** Full marks if and only if all correct choices are marked and no incorrect choice is selected. Zero marks for partial selection or any incorrect choice. No negative marking.
- **Numerical Answer Type (NAT):** Any numerical value falling strictly within the specified official range (e.g., [a, b]) receives full credit. No negative marking.
- **Marks to All (MTA):** Questions officially designated as MTA award full marks to all candidates regardless of attempt.